

AWS EC2 Free Tier Deployment Guide for Flask Machine Learning Application

Objective: Deploy a Flask Machine Learning application on an AWS EC2 Ubuntu instance.

Prerequisites

Before starting, ensure you have:

- AWS Account
 - Flask Application (`app.py`)
 - Trained ML model (`model.pkl`)
 - HTML templates
 - Internet connection
 - SSH Key Pair (`.pem`)
 - WinSCP (for file transfer)
-

Deployment Workflow

```
Create AWS Account
      ↓
Launch EC2 Instance
      ↓
Configure Security Group
      ↓
Connect using SSH
      ↓
Install Python & Dependencies
      ↓
Transfer Project Files
      ↓
Run Flask Application
      ↓
Access Website using Public IP
```

Step 1: Create AWS Account

Navigation

```
AWS Console
├── Create Account
├── Billing Information
├── Phone Verification
└── Login
```

Step 2: Launch EC2 Instance

Navigation

```
AWS Console
  ↓
Services
  ↓
EC2
  ↓
Launch Instance
```

Configuration

Instance Name

Flask-Deployment

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS

Instance Type

t3.micro (Free Tier Eligible)

Key Pair

Create a new key pair.

Example

deployment-key

Download

deployment-key.pem

Store this file securely.

Step 3: Configure Network

Click **Edit** under Network Settings.

Enable:

- SSH (22)
- HTTP (80)

Add one more inbound rule.

Type	Port	Source
Custom TCP	5000	Anywhere

Finally,

Launch Instance

Step 4: Connect to EC2

Navigation

EC2



Instances



Select Instance



Connect



SSH Client

Copy the generated command.

Example

```
ssh -i deployment-key.pem ubuntu@ec2-xx-xx-xx-xx.eu-north-1.compute.amazonaws.com
```

If prompted,

yes

Step 5: Update Ubuntu

```
sudo apt update
```

```
sudo apt upgrade -y
```

Step 6: Install Python

```
sudo apt install python3-pip -y
```

Verify installation.

```
python3 --version
```

```
pip3 --version
```

Step 7: Install Required Python Libraries

Install individually

```
pip3 install flask
```

```
pip3 install numpy
```

```
pip3 install pandas
```

```
pip3 install scikit-learn
```

or

```
pip3 install flask numpy pandas scikit-learn
```

Step 8: Install WinSCP

Install WinSCP on Windows.

Open WinSCP.

Provide

Field	Value
File Protocol	SFTP
Host Name	Public IPv4 DNS
User Name	ubuntu
Port	22
Private Key	deployment-key.ppk

Click **Login**.

Step 9: Upload Project

Upload the following files.

```
app.py
model.pkl
templates/
static/
requirements.txt
```

Step 10: Verify Uploaded Files

ls

Expected output

```
app.py
model.pkl
templates
static
requirements.txt
```

Step 11: Modify Flask Application

Replace

```
app.run(debug=True)
with
```

```
app.run(host="0.0.0.0", port=5000)
```

This allows external users to access the application.

Step 12: Run Flask

```
python3 app.py
```

Expected output

Running on http://0.0.0.0:5000

Step 13: Find Public IP Address

Navigation

EC2



Instances



Select Instance



Public IPv4 Address

Example

13.xx.xx.xx

Open browser.

http://13.xx.xx.xx:5000

Your application should open.

Step 14: If Website Does Not Open

Navigate to

EC2



Security Groups



Inbound Rules

Add

Type	Port	Source
Custom TCP	5000	0.0.0.0/0

Save changes.

Refresh browser.

Frequently Used Linux Commands

Current Directory

pwd

List Files

```
ls
```

Change Directory

```
cd folder_name
```

Go Back

```
cd ..
```

Create Folder

```
mkdir project
```

Delete File

```
rm filename
```

Clear Terminal

```
clear
```

Install Everything Together

```
sudo apt update
```

```
sudo apt install python3-pip -y
```

```
pip3 install flask numpy pandas scikit-learn
```

Project Structure

```
Deployment/  
|  
├── app.py  
├── model.pkl  
├── requirements.txt  
└── templates/
```

```
└─ index.html
└─ static/
```

Mathematical Representation of Deployment Flow

Let

- (A) = AWS Account
- (E) = EC2 Instance
- (P) = Python Environment
- (D) = Dependencies
- (F) = Flask Application
- (M) = Machine Learning Model

The deployment pipeline can be represented as

[A $\rightarrow$ E $\rightarrow$ P $\rightarrow$ D $\rightarrow$ F $\rightarrow$ M $\rightarrow$ Public Deployment]

Network Communication

The client accesses the deployed application as

[Client Browser $\rightarrow$ $\text{EC2 Public IP} : 5000$ $\rightarrow$ Flask Application]

Summary

1. Create AWS Account
2. Launch EC2 Instance
3. Configure Security Group
4. Connect via SSH
5. Update Ubuntu
6. Install Python
7. Install Dependencies
8. Upload Project
9. Run Flask
10. Access via Public IP

In []: